Spectral Radius Minimization for Signed Squares of Cycles

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Abstract

For even $n \geq 8$, let $C_n^2 = C_n(1,2)$ be the square of the cycle, with its distance-one and distance-two edges independently assigned signs, with no periodicity assumption on the signing, and let m_n be the minimum signed adjacency spectral radius. A distinguished twisted signing attains $\rho_-(n)$, where $\rho_-(n)^2 = 4 + 2\cos(2\pi/n) + 2\cos(4\pi/n)$, at every even order. We prove that equality holds exactly for $n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}$; equivalently, strict failure occurs precisely at 32, at 40, and at every even $n \geq 48$. Structurally, a period-eight reference phase has squared spectral edge $\eta < 8$, while its six-gap phase slip is the unique least-edge abnormal positive single gap and has an isolated rank-two squared edge $c_6 \in (\eta, 8)$. Charge-compatible separated copies of this slip, together with a discrete IMS estimate, prove eventual failure, while exact finite verification shows that continuous failure begins precisely at order 48. The finite portions are reduced to explicitly finite objects and independently verified using exact arithmetic, so floating-point evidence makes no spectral decision.

Keywords: signed graph; spectral radius; cycle square; switching equivalence; phase slip; computer-assisted proof

MSC 2020: 05C50 (primary); 05C22, 68V05 (secondary)

1 Introduction

Let $n \geq 8$ be even. We write

$$G_n = C_n^2 = C_n(1,2)$$

for the graph on $\mathbb{Z}/n\mathbb{Z}$ in which two vertices are adjacent when their cyclic distance is one or two. An edge signing is a map $\sigma : E(G_n) \to \{\pm 1\}$, and A_σ is the real symmetric matrix whose i, j entry is $\sigma(ij)$ on an edge and zero otherwise. Our problem is to determine

$$m_n = \min_{\sigma : E(G_n) \to \{\pm 1\}} \rho(A_\sigma).$$

Thus the already formed cycle square is being signed: the distance-one and distance-two edges receive signs independently. It is not the graph obtained by first signing a cycle and then squaring that signed object. Switching at vertices conjugates A_σ by a diagonal $\{\pm 1\}$-matrix, so the minimization naturally takes place over switching classes.

The problem belongs to the general signing program in spectral graph theory. Signings encode the new eigenvalues of two-lifts in the work of Bilu and Linial [3], while Belardo, Cioabă, Koolen, and Wang ask explicitly which signatures minimize the adjacency spectral radius of a fixed connected graph [2]. Interlacing families give strong existence results for one side of the signed spectrum; in the bipartite setting spectral symmetry turns this into the two-sided control used to construct Ramanujan lifts [9]. The graphs C_n^2 are nonbipartite, however, so a one-sided eigenvalue estimate does not by itself control $\rho(A_\sigma)$. Here the graph is fixed at each order and the question is the exact two-sided optimum, rather than an existence bound uniform over a broad graph class.

The problem-specific starting point is Suvagiya's preprint [10, Conjecture 3]. For $C_n(1,2)$, it identifies four distinguished switching classes in which every quadrilateral is unbalanced, computes the spectra of the two twisted classes, verifies global optimality for even orders through 18, and formulates the all-even-order assertion as Conjecture 3. We determine exactly when that twisted candidate is globally minimizing, thereby resolving and disproving the conjecture. In particular, the problem, the flux coordinates, and the twisted spectral formula are inputs from that work, not novelty claims of the present paper.

The answer is unexpectedly nonmonotone. The proposed equality holds for every even order from 8 through 30, fails first at 32, returns at $34, 36, 38$, fails again at 40, and returns once more at $42, 44, 46$. Only from 48 onward does failure persist at every even order. The first counterexample therefore does not mark the beginning of continuous failure, and the two isolated failures cannot be inferred from the eventual phase-slip mechanism developed below.

To state the classification, set

$$\rho_-(n)^2 = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n}, \qquad \theta_n = \rho_-(n)^2.$$

The distinguished class has alternating triangle flux and negative Hamilton-cycle holonomy; a direct antiperiodic Fourier calculation shows that it attains $\rho_-(n)$ for every even $n \geq 8$. This gives the upper bound $m_n \leq \rho_-(n)$ at all orders. At an order where equality holds, the reverse inequality for every signing is a separate conclusion of exact exhaustion.

Theorem 1.1 (Complete classification)**.** *For every even $n \geq 8$, the distinguished twisted signing attains $\rho_-(n)$, and*

$$m_n < \rho_-(n) \quad \iff \quad n = 32, \quad n = 40, \quad \text{or} \quad n \geq 48.$$

Equivalently,

$$m_n = \rho_-(n) \quad \iff \quad n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}.$$

Theorem 1.1 classifies the truth of the proposed equality. It neither classifies all minimizing signings at the valid orders nor determines the exact value of m_n at the failing orders.

Its finite component determines the irregular small-order truth table. A different part of the argument explains why failure is eventually uninterrupted: a low-spectral periodic phase supports localized elementary interfaces, and well-separated copies of those interfaces can be closed on large rings. The assertion that continuous failure begins at 48 uses both this analytic tail and an exact finite bridge; the interface theory alone gives no such sharp onset.

The structural comparison phase is period eight and is distinct from the twisted signing that attains $\rho_-(n)$. In its quadrilateral-flux word, the positive sites occur at spacing four. Replacing one spacing by a positive gap g creates a single interface; we call $q = g - 4$ its excess charge. For a cyclic collection of gaps, the global closing law is $\sum_j q_j \equiv n \pmod 8$, whereas the local translation sector carried by one charge is $\sigma_{\mathrm{sec}}(q) = q \pmod 4$. These congruences have different roles and are not interchangeable. In particular, one, two, and three copies of the gap-six interface, whose charge is two, supply the three nonzero even residue classes needed in the finite-ring constructions.

The reference squared spectral edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

For the bilateral positive single-gap operator A_g, put $H_g = A_g^2$ and $E(g) = \sup \operatorname{Spec}(H_g)$. Define c_6 to be the unique root in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}$$

of the degree-ten polynomial

$$\begin{aligned}p_6(y) = 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ - 19968y + 256.\end{aligned}$$

The exact isolation is important: all later strict comparisons are made against rational endpoints, not against a decimal approximation.

Theorem 1.2 (Reference and single-gap spectral hierarchy). *For both lifts and both orientations,*

$$E(4) = \eta < c_6 = E(6), \qquad \mathrm{Spec}_{\mathrm{ess}}(H_6) = \mathrm{Spec}(H_{\mathrm{ref}}), \qquad \dim\ker(H_6 - c_6) = 2.$$

Moreover, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

The scope of Theorem 1.2 is precise: G6 is the unique minimizer among abnormal positive single gaps. It is not asserted to minimize over arbitrary multi-gap or finite-core interfaces. The strict single-gap hierarchy makes G6 the elementary low-cost defect above the period-eight bulk. Its charge allows one, two, or three separated copies to close in the required residue classes. Exact patch identification transfers the bilateral reference/G6 bounds to those finite rings, and a discrete IMS partition controls the squared spectral top between the patches. This proves failure for every even $n \geq 240$. Exact finite verification on $48 \leq n < 240$ then shows that continuous failure begins precisely at 48.

The computer-assisted part is organized as a proof by finite reduction, not as numerical evidence. Switching quotients and local interlacing reduce the relevant cases to finite exact objects. For $8 \leq n \leq 32$, complete switching-class decisions establish the lower bounds through 30 and an exact witness at 32. For $34, 36, 38, 42, 44, 46$, a parity-lifted finite state graph is proved sound, complete, and terminating, and every terminal case is closed exactly; a separate rational positive-definiteness certificate gives the witness at 40. Finally, rational positive-definiteness certificates cover every even order from 48 through 238. Independent checkers reconstruct the finite objects and repeat the integer, rational, or algebraic decisions. Exhaustive generation and local finite-state reductions have established precedents in graph theory [7, 5], while all-order classifications often combine an analytic tail with finite closure [8, 6].

Exact signed spectral-radius classifications are known in settings where the underlying graph is also allowed to vary, including unbalanced unicyclic and bicyclic graphs [1] and connected unbalanced signed graphs of fixed order [4]. Those results make it essential to keep the present novelty claim narrow. To the best of our knowledge, no previous work gives an all-order classification of the minimizing behavior over all real edge signings of the fixed circulant family $C_n(1, 2)$. The closest direct source remains Suvagiya's conjecture; the complete truth set and the elementary single-gap mechanism are the two contributions supplied here. The paper is arranged as follows. Section 2 sets up switching coordinates, the twisted attaining candidate, and the period-eight reference phase. Section 3 develops gaps, charges, and the two closing laws. Sections 4 and 5 prove the G6 edge theorem and the complete single-gap hierarchy. Section 6 passes from bilateral interfaces to finite rings by patch identification and discrete IMS localization. Section 7 closes the remaining orders by exact finite arguments, and Section 8 records the resulting perspective. The appendices collect the algebraic G6 certificate and the exact finite-classification boundary.

2 Switching Coordinates and the Reference Phase

2.1 Edge signings, switching, and Hamilton gauge

Write the vertices of $G_n = C_n(1,2)$ as $\mathbb{Z}/n\mathbb{Z}$. Let a_i be the sign of the distance-one edge $\{i, i+1\}$, and let b_i be the sign of the distance-two edge $\{i, i+2\}$, with all subscripts read cyclically. Switching by $\varepsilon = (\varepsilon_i)_{i\in\mathbb{Z}/n\mathbb{Z}} \in \{\pm 1\}^n$ changes these signs to

$$a'_i = \varepsilon_i a_i \varepsilon_{i+1}, \qquad b'_i = \varepsilon_i b_i \varepsilon_{i+2}.$$

At the matrix level this is the orthogonal conjugacy $A_{\sigma'} = D_\varepsilon A_\sigma D_\varepsilon$, where $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$. Switching therefore preserves the complete spectrum, not only the spectral radius.

Lemma 2.1 (Hamilton gauge). *Every signing of G_n is switching equivalent to one for which*

$$a_0 = \cdots = a_{n-2} = 1, \qquad a_{n-1} = \alpha,$$

where

$$\alpha = \prod_{i=0}^{n-1} a_i \in \{\pm 1\}.$$

The residual sign α, the Hamilton-cycle holonomy, is invariant under switching.

Proof. Set $\varepsilon_0 = 1$ and recursively choose $\varepsilon_{i+1} = \varepsilon_i a_i$ for $0 \le i \le n-2$. Then $a'_i = 1$ along the spanning path $0, 1, \ldots, n-1$. On the closing edge,

$$a'_{n-1} = \varepsilon_{n-1} a_{n-1} \varepsilon_0 = \prod_{i=0}^{n-1} a_i.$$

The product of the signs around a cycle is unchanged by switching because every vertex factor occurs twice. □

2.2 Flux words, holonomy, and the attaining candidate

Before fixing a gauge, define the triangle flux at i by

$$\tau_i = a_i a_{i+1} b_i.$$

Each switching factor again occurs twice, so τ_i is invariant. In the Hamilton gauge of Lemma 2.1, the actual distance-two signs are

$$b_i = \tau_i \quad (0 \le i \le n-3), \qquad b_{n-2} = \alpha\tau_{n-2}, \qquad b_{n-1} = \alpha\tau_{n-1}.$$

Consequently (τ, α) parametrizes the switching classes bijectively once the vertex labels and seam are fixed. It is often cleaner to absorb the closing sign into the boundary condition. More precisely, let

$$\mathcal{U}_n^\alpha = \{u : \mathbb{Z} \to \mathbb{C} : u_{j+n} = \alpha u_j\}.$$

On this n-dimensional space the signed adjacency takes the form

$$(A_\tau u)_i = u_{i-1} + u_{i+1} + \tau_{i-2} u_{i-2} + \tau_i u_{i+2}. \tag{1}$$

This twisted-boundary realization includes every edge that crosses the Hamilton-gauge seam and is unitarily identical to the original finite signed matrix. We suppress α from A_τ when the boundary space $\mathcal{U}_n^\alpha$ is displayed explicitly.

The quadrilateral-flux word is

$$Q_i = \tau_i \tau_{i+1}.$$

It satisfies $\prod_i Q_i = 1$. Conversely, a cyclic word $Q \in \{\pm 1\}^n$ has a cyclic τ-lift precisely when $\prod_i Q_i = 1$; after choosing τ_0, the recurrence

$$\tau_{i+1} = Q_i \tau_i$$

determines that lift. The two choices of τ_0 give τ and $-\tau$, while α remains independent. Thus the exact cyclic data are (τ, α), or equivalently (Q, τ_0, α). The reduced pair (Q, α) represents two switching classes, a distinction that disappears only after passing to the squared spectrum or spectral radius. The lift condition $\prod_i Q_i = 1$ and the Hamilton twist α are independent. If n is even and $(Du)_i = (-1)^i u_i$, then

$$A_{-\tau} = -DA_\tau D, \qquad H_{-\tau} = DH_\tau D, \qquad H_\tau := A_\tau^2. \tag{2}$$

The unsquared spectra of the two lifts are reflected through zero; their squared spectra and spectral radii agree.

Proposition 2.2 (Attainment by the twisted candidate)**.** *For every even $n \geq 8$, the class*

$$Q_i = -1 \quad (0 \leq i < n), \qquad \alpha = -1,$$

has a representative $\tau_i = (-1)^i$ satisfying

$$\rho(A_\tau)^2 = 4\cos^2\frac{\pi}{n} + 4\cos^2\frac{2\pi}{n} = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n} = \theta_n.$$

Consequently $m_n \leq \rho_-(n)$ for every even $n \geq 8$.

Proof. Because $\alpha = -1$, the allowed Fourier parameters on $\mathcal{U}_n^{-1}$ are

$$\vartheta_k = \frac{(2k+1)\pi}{n}, \qquad 0 \leq k < n.$$

Multiplication by $(-1)^j$ carries the mode $e^{ij\vartheta}$ to $e^{ij(\vartheta+\pi)}$. Hence the span of these two modes is invariant, and in the displayed order the restriction of A_τ is

$$B(\vartheta) = 2\begin{pmatrix} \cos\vartheta & \cos 2\vartheta \\ \cos 2\vartheta & -\cos\vartheta \end{pmatrix}.$$

Thus

$$B(\vartheta)^2 = 4(\cos^2\vartheta + \cos^2 2\vartheta) I_2.$$

With $x = \cos 2\vartheta$, the quantity in parentheses is

$$F(x) = x^2 + \frac{x}{2} + \frac{1}{2}.$$

The relevant values lie in $[-1, \cos(2\pi/n)]$. Since F is convex, its maximum is at an endpoint. Now $F(-1) = 1$, whereas $n \geq 8$ gives

$$F\left(\cos\frac{2\pi}{n}\right) \geq F\left(\frac{1}{\sqrt{2}}\right) = 1 + \frac{1}{2\sqrt{2}} > 1.$$

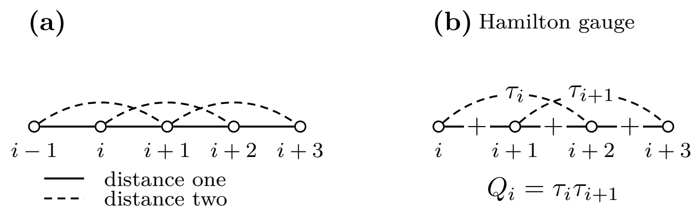

Figure 1: Switching coordinates for the cycle square. Distance-one and distance-two edges are signed independently; after Hamilton gauge, the local triangle word τ and its adjacent products $Q_i = \tau_i\tau_{i+1}$ record the flux data.

The maximum therefore occurs at $\vartheta = \pi/n$ or $\pi - \pi/n$, which yields the first equality. The double-angle identity yields the second. Since this explicit signing is in the minimization domain, the final inequality follows. $\square$

The candidate in Proposition 2.2 is the signing that attains the conjectured comparison value. It is not the period-eight reference phase introduced in Subsection 2.4. The latter is a different low-spectral bulk used to construct phase slips.

2.3 The squared local operator

For a bilateral word τ, equation (1) defines a bounded self-adjoint range-two operator on $\ell^2(\mathbb{Z})$. Its square $H_\tau = A_\tau^2$ is the nonnegative range-four operator

$$\begin{aligned}(H_\tau u)_i = {}& 4u_i + u_{i-2} + u_{i+2} \\ &+ \tau_{i-4}\tau_{i-2}u_{i-4} + (\tau_{i-3} + \tau_{i-2})u_{i-3} \\ &+ (\tau_{i-2} + \tau_{i-1})u_{i-1} + (\tau_{i-1} + \tau_i)u_{i+1} \\ &+ (\tau_i + \tau_{i+1})u_{i+3} + \tau_i\tau_{i+2}u_{i+4}. \end{aligned} \tag{3}$$

Indeed, this follows by inserting equation (1) four times into $A_\tau(A_\tau u)$ and collecting equal displacements. The extreme coefficients may also be written

$$\tau_{i-4}\tau_{i-2} = Q_{i-4}Q_{i-3}, \qquad \tau_i\tau_{i+2} = Q_iQ_{i+1},$$

while, for example, $\tau_i + \tau_{i+1} = \tau_i(1 + Q_i)$. Thus a negative Q_i cancels the associated odd-displacement coupling. Formula (3) is local and precedes every periodic or interface specialization.

Translation of the coefficient word is implemented by the bilateral shift. Reflection reverses the word and is unitary after the corresponding index change. Equation (2) handles the two lifts. We will use only these explicit equivalences in the body; the fiber-level zone-folding identities are recorded with the Floquet algebra in Appendix A.

2.4 The period-eight reference phase

The canonical reference word and its quadrilateral flux are

$$\tau_{\mathrm{ref}} = (+,+,-,+,-,-,+,-), \qquad Q_{\mathrm{ref}} = (+,-,-,-,+,-,-,-),$$

extended periodically. The positive Q_{ref}-sites are exactly $4\mathbb{Z}$. More generally, for $s \in \mathbb{Z}/4\mathbb{Z}$, let B_s denote the translate whose positive Q-sites are $s + 4\mathbb{Z}$. Each B_s has two τ-lifts; since one four-site cell has Q-product -1, those lifts are antiperiodic over four sites and periodic over eight.

For a nonzero Bloch multiplier z, impose

$$u_{8m+r} = z^m v_r, \qquad 0 \le r < 8,$$

and write $A_{\mathrm{ref}}(z)$ for the resulting unsquared 8×8 fiber. It is Hermitian when $|z| = 1$.

Proposition 2.3 (Reference squared edge)**.** *Let*

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

Then

$$\sup_{|z|=1} \rho\big(A_{\mathrm{ref}}(z)\big)^2 = \eta < 8,$$

and equality occurs only at $z = 1$.

Proof. The exact determinant identity is derived and independently certified in Appendix A. We retain here the two algebraic steps that locate the edge. At $z = 1$, put $y = x^2$ and $X = y - 4$. The fiber equation reduces to

$$X^4 - 20X^2 + 80 = 0.$$

Its largest y-root is $4 + \sqrt{10 + 2\sqrt{5}} = \eta$.

For global maximality, set

$$s = \sqrt{10 + 2\sqrt{5}}, \qquad u = y - \eta, \qquad t = 2 - (z + z^{-1}).$$

On the unit circle, $t \in [0, 4]$. The same exact determinant relation, after substitution, is

$$\begin{aligned} 0 = {} & u^4 + 4su^3 + 2u^2t + (40 + 12\sqrt{5})u^2 + 4sut + 8\sqrt{5}\, su \\ & + t^2 + (4\sqrt{5} - 3)t. \end{aligned}$$

Every displayed coefficient is positive. If a squared fiber eigenvalue satisfied $y \ge \eta$, then $u, t \ge 0$, so the right-hand side could vanish only for $u = t = 0$. Thus $y \le \eta$, and equality forces $z + z^{-1} = 2$, hence $z = 1$. Finally, $\sqrt{5} < 9/4$ and $\sqrt{10 + 2\sqrt{5}} < 191/50$ give $\eta < 391/50 < 8$. □

The reference phase is also the zero-interface member of the gap language. If consecutive positive Q-sites are separated by $g = 4$, the right positive lattice is the continuation of the left lattice in the same translate B_s. Up to translation, reflection, and the lift choice, this is precisely the unperturbed reference bulk, not an abnormal interface.

3 Gaps, Charges, and Translation Sectors

3.1 Cyclic gaps and excess charge

Let $Q \in \{\pm 1\}^n$ be a cyclic quadrilateral-flux word that admits a cyclic τ-lift. Suppose first that its positive set

$$D(Q) = \{i \in \mathbb{Z}/n\mathbb{Z} : Q_i = +1\}$$

is nonempty. Choose integer representatives

$$x_1 < x_2 < \cdots < x_d < x_1 + n$$

for its d elements and put $x_{d+1} = x_1 + n$. The positive cyclic gaps and their excess charges are

$$g_j = x_{j+1} - x_j, \qquad q_j = g_j - 4 \qquad (1 \le j \le d).$$

The reference value is four; positive or negative q_j measures the departure of one gap from that spacing.

Proposition 3.1 (Gap and charge identities)**.** *The gap list satisfies*

$$\sum_{j=1}^{d} g_j = n, \qquad \sum_{j=1}^{d} q_j = n - 4d.$$

If n is even and Q has a cyclic τ-lift, then d is even and

$$\sum_{j=1}^{d} q_j \equiv n \pmod 8.$$

Proof. The half-open arcs from x_j to x_{j+1} partition one traversal of the n-cycle, so their lengths sum to n. Subtracting four for each of the d arcs gives the second identity.

The word has d positive and $n-d$ negative entries. Hence

$$\prod_{i=0}^{n-1} Q_i = (-1)^{n-d}.$$

A cyclic lift exists only when this product is 1. Thus $n-d$ is even; since n is even, d is even. Write $d = 2h$. The second identity becomes

$$\sum_{j=1}^{d} q_j = n - 8h,$$

which is the asserted congruence. $\square$

The Hamilton holonomy α does not occur in this argument. It is an independent step-one cycle sign and cannot change the charge law. If $D(Q) = \varnothing$, there is no cyclic gap list and Proposition 3.1 is not invoked; this defect-free word is treated separately whenever it occurs.

3.2 Translation sectors

Recall the four reference translates B_s, $s \in \mathbb{Z}/4\mathbb{Z}$, defined by

$$(B_s)_i = +1 \iff i \equiv s \pmod 4.$$

They are sectors of the Q-word. Each has two period-eight τ-lifts, but the sector label itself depends only on the positions of the positive Q-sites.

Proposition 3.2 (Oriented gap shift)**.** *Suppose an oriented gap of length g leaves the reference sector B_s at a positive site and enters a reference sector on its right. For its charge $q = g - 4$, the right sector is*

$$B_{s+\sigma_{\mathrm{sec}}(q)}, \qquad \sigma_{\mathrm{sec}}(q) = q \pmod 4.$$

This law is independent of the τ-lift and Hamilton holonomy.

Proof. Let the left endpoint be $x \equiv s \pmod 4$. The next positive site is $x + g$, so the positive lattice continuing to the right is

$$x + g + 4\mathbb{Z}.$$

It is therefore B_{s+g}. Since $g = q + 4$, the residues of g and q agree modulo four. The proof uses only endpoint positions, so neither the lift sign nor α enters. $\square$

3.3 Composition and legal cyclic closure

For consecutive oriented gaps with charges $q_1, \ldots, q_k$, repeated use of Proposition 3.2 gives

$$B_s \longmapsto B_{s+q_1} \longmapsto \cdots \longmapsto B_{s+q_1+\cdots+q_k},$$

with labels reduced modulo four. Thus sector shifts add:

$$\sigma_{\mathrm{sec}}(q_1 + \cdots + q_k) \equiv \sum_{j=1}^{k} \sigma_{\mathrm{sec}}(q_j) \pmod 4.$$

On a cyclic gap list, $\sum_j g_j = n$, so the final reference lattice is the translate B_{s+n}; equivalently,

$$\sum_j q_j \equiv n \pmod 4.$$

This compatibility is weaker than the mod-eight law in Proposition 3.1, because it does not contain the parity information needed for a cyclic τ-lift.

The modulo-eight charge closure and the modulo-four translation-sector law encode different constraints and will be used separately. The first is a global condition on a liftable ring; the second records the local reference translate across an interface.

For later use, a gap of length six has charge $q = 6-4 = 2$ and sector shift two modulo four. One, two, or three such gaps contribute total charges $2, 4, 6$, respectively, and sector shifts $2, 0, 2$ modulo four. If all other gaps have the reference length four, these are precisely the charge contributions required for ring orders congruent to $2, 4, 6$ modulo eight. No spectral comparison is asserted at this stage.

4 The Elementary Six-Gap Phase Slip

4.1 The bilateral interface and its essential spectrum

The elementary interface replaces one reference gap of length four by a gap of length six. On $\mathbb{Z}$, let

$$D_6 = \{-4j : j \geq 0\} \cup \{6 + 4j : j \geq 0\}$$

and set $Q_i^{(6)} = +1$ exactly for $i \in D_6$. Choose the lift $\tau_0^{(6)} = 1$ and $\tau_{i+1}^{(6)} = Q_i^{(6)} \tau_i^{(6)}$. We write

$$(A_6 u)_i = u_{i-1} + u_{i+1} + \tau_{i-2}^{(6)} u_{i-2} + \tau_i^{(6)} u_{i+2}, \qquad H_6 = A_6^2.$$

The other lift has the same squared spectrum by equation (2); reflection gives the opposite interface orientation. The left tail has positive Q-sites $4\mathbb{Z}$, while the right tail has positive sites $6 + 4\mathbb{Z}$. They are translates B_0 and B_2 of the reference phase.

Every row of A_6 has four entries of absolute value one, and the matrix is real symmetric. Hence A_6 is bounded and self-adjoint, $\|A_6\| \leq 4$, and

$$0 \leq H_6 \leq 16I.$$

Lemma 4.1 (Essential spectrum and tail matching)**.** *The G6 squared operator satisfies*

$$\mathrm{Spec}_{\mathrm{ess}}(H_6) = \mathrm{Spec}(H_{\mathrm{ref}}), \qquad \sup \mathrm{Spec}_{\mathrm{ess}}(H_6) = \eta.$$

Every $y > \eta$ in $\mathrm{Spec}(H_6)$ *is an isolated eigenvalue of finite multiplicity. Its nonzero unsquared components decay exponentially on both periodic tails and obey the stable/unstable matching criterion in Subsection 4.2.*

Proof. Choose two cuts outside the finite interface core. Decoupling at those cuts gives the direct sum of a left periodic half-line compression, a finite middle block, and a right periodic half-line compression. Since H_6 has range four, its difference from this decoupled operator has finite rank. The finite block contributes no essential spectrum.

For completeness, a periodic half-line compression has the same essential spectrum as its whole-line operator. In one direction, a whole-line Bloch solution cut off on L cells far down the half-line has norm of order $L^{1/2}$, while the residual is confined to a fixed number of boundary sites. Normalization and Weyl's criterion put every whole-line spectral point in the half-line essential spectrum. Conversely, if x lies in the whole-line resolvent and $R = (H_{\mathrm{per}} - x)^{-1}$, then the compression PRP is a two-sided inverse for the half-line operator modulo finite-rank cross-boundary terms. The half-line operator is therefore Fredholm at x. This proves the reverse inclusion.

Both whole-line tail operators are unitarily equivalent to H_{ref} by translation and diagonal switching. Finite-rank invariance now proves the displayed essential-spectrum identity, and Proposition 2.3 supplies its upper edge. Standard self-adjoint spectral theory then makes every spectral point above η discrete with finite multiplicity.

Let $H_6 u = yu$, $y > \eta$, and $\lambda = \sqrt{y}$. The orthogonal branch decomposition is

$$u_\pm = \frac{1}{2}\left(u \pm \lambda^{-1} A_6 u\right), \qquad A_6 u_\pm = \pm\lambda u_\pm.$$

At least one branch is nonzero. On either periodic tail the fourth-order recurrence has a four-dimensional eight-step monodromy. A unit-circle multiplier would give a reference Bloch state at squared energy $y > \eta$, contradicting Proposition 2.3. The reciprocal multiplier structure therefore gives two multipliers inside and two outside the unit circle. An ℓ^2 branch lies in the left unstable plane and the right stable plane; iteration gives exponential decay. The finitely many transfer maps within one period preserve the cell-state decay at individual sites. Invertibility of the recurrence proves the converse matching assertion. □

4.2 Algebraic candidate and physical matching

Define

$$\begin{aligned} p_6(y) = {} & 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ & - 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ & - 19968y + 256 \end{aligned}$$

and let c_6 be the unique root of p_6 in

$$J_6 = \left(\frac{7905369311620327}{10^{15}}, \frac{7905369311620328}{10^{15}}\right). \tag{4}$$

Thus $c_6 \approx 7.905369311620327$, but the rational interval, not this decimal, is used in every proof.

Fix $y > \eta$ and first take the positive branch $\lambda = \sqrt{y}$. At a left bulk cut let $U_L(\lambda)$ be the two-plane of transfer data decaying toward $-\infty$; at a right cut let $S_R(\lambda)$ be the two-plane decaying toward $+\infty$. If $D_6(\lambda)$ is transfer through the finite interface core, then

$$y \text{ is a positive-branch physical level} \quad \Longleftrightarrow \quad D_6(\lambda)U_L(\lambda) \cap S_R(\lambda) \neq \{0\}.$$

For any bases u_1, u_2 of U_L and s_1, s_2 of S_R, this is equivalent to

$$E_6(\lambda) = \det[D_6(\lambda)u_1, D_6(\lambda)u_2, s_1, s_2] = 0. \tag{5}$$

A basis change multiplies this determinant by a nonzero factor. Its zero set is therefore intrinsic; Grassmann charts are only coordinate representatives of this geometric condition.

Lemma 4.2 (Exact physical realization)**.** *The positive-branch determinant has exactly one zero over J_6; it is simple and its square is c_6. Consequently*

$$+\sqrt{c_6} \in \mathrm{Spec}(A_6), \qquad c_6 \in \mathrm{Spec}(H_6).$$

Proof. On J_6, the two stable multipliers are real, distinct, and uniformly separated from the unit circle. An exact rational interval evaluation of the genuine determinant (5) has opposite endpoint signs, and its derivative has a fixed nonzero sign throughout the interval. Hence there is exactly one simple physical positive root. At that root the matching intersection is one-dimensional: if its dimension were at least two, the 4×4 matching matrix would have rank at most two, its adjugate would vanish, and the derivative of its determinant would also vanish. Invertibility of the transfer recurrence then identifies this one-dimensional intersection with the positive eigenspace.

To identify its square, symmetric coordinates for the two stable multipliers reduce the bulk equations and the unsquared matching equation to a finite polynomial system. Exact elimination contains $p_6(y)^2$, and Sturm isolation proves that p_6 has exactly one root in J_6. Since physical existence and uniqueness were established before elimination, the squared physical root must be c_6. The interval enclosures, derivative bound, elimination identity, and Sturm count are given in Appendix A. In particular, an elimination root is not declared physical merely because it is a root of a resultant. □

4.3 The global G6 edge

It remains to prove that no other physical level has greater absolute unsquared value. The local argument above excludes another positive root through the upper endpoint of J_6. On the remaining compact energy interval up to the row-sum bound 16, the stable and unstable planes are covered by a finite exact Grassmann atlas. Exact elimination and Sturm counts give a complete finite list of algebraic candidate intervals. On every such interval the genuine unsquared determinant is bounded away from zero; a second coordinate reconstruction gives the same exclusions. The atlas also covers its chart transitions and the sole repeated multiplier, which lies away from the unit circle. These exact finite statements are proved in Appendix A; no floating-point sign is used.

Proposition 4.3 (Positive-branch completeness)**.** *The eigenvalue $+\sqrt{c_6}$ is simple, and A_6 has no positive eigenvalue larger than $\sqrt{c_6}$.*

Proof. Lemma 4.2 gives existence and simplicity. Lemma 4.1 reduces every other positive spectral point above η to the physical matching determinant. The complete exact candidate audit just described excludes all its zeros above c_6, and $\|A_6\| \leq 4$ closes the squared interval above 16. □

The symmetry proved in the next subsection reflects every negative unsquared level to a positive one. Combining it with Proposition 4.3 and Lemma 4.2 yields, for either lift and orientation,

$$\sup \mathrm{Spec}(H_6) = c_6. \tag{6}$$

4.4 Rank-two symmetry

The G6 word has the all-integer reflection identities

$$Q^{(6)}_{6-i} = Q^{(6)}_i, \qquad \tau^{(6)}_{7-i} = -\tau^{(6)}_i \qquad (i \in \mathbb{Z}). \tag{7}$$

The first follows directly from the two reflected positive lattices in D_6. For the second, one checks the anchor $\tau^{(6)}_7 = -\tau^{(6)}_0$. If it holds at i, then

$$\tau^{(6)}_{6-i} = Q^{(6)}_{6-i}\tau^{(6)}_{7-i} = -Q^{(6)}_i\tau^{(6)}_i = -\tau^{(6)}_{i+1}.$$

This propagates the identity in one direction, and the same recurrence backwards proves it for every integer.

Define the unitary operator

$$(Ku)_i = (-1)^i u_{9-i}.$$

Applying it twice and substituting equation (7) into the four terms of A_6 gives

$$K^2 = -I, \qquad KA_6 = -A_6K, \qquad KH_6 = H_6K.$$

Thus K maps the simple eigenspace at $+\sqrt{c_6}$ bijectively to a simple eigenspace at $-\sqrt{c_6}$, and it maps any hypothetical negative level of larger absolute value to an excluded positive level. This proves the spectral reflection used in equation (6).

Finally, self-adjointness gives the orthogonal decomposition

$$\ker(H_6 - c_6) = \ker(A_6 - \sqrt{c_6}) \oplus \ker(A_6 + \sqrt{c_6}).$$

Both summands are one-dimensional, so

$$\dim \ker(H_6 - c_6) = 2.$$

The squared level c_6 is therefore rank two; it is not simple. The two unsquared partners $+\sqrt{c_6}$ and $-\sqrt{c_6}$ are simple and their eigenvectors are exponentially localized by Lemma 4.1.

5 Optimality Among Single-Gap Interfaces

5.1 Canonical single-gap operators

For a positive integer g, let

$$D_g = \{-4j : j \geq 0\} \cup \{g + 4j : j \geq 0\}.$$

Define $Q_i^{(g)} = +1$ exactly on D_g, choose the lift $\tau_0^{(g)} = 1$, and set

$$(A_g v)_i = v_{i-1} + v_{i+1} + \tau_{i-2}^{(g)} v_{i-2} + \tau_i^{(g)} v_{i+2}, \qquad H_g = A_g^2, \qquad E(g) = \sup \operatorname{Spec}(H_g).$$

Reflection covers the reverse orientation, and equation (2) covers the other τ-lift. It is therefore enough to make each comparison for the displayed forward lift.

Gap $g = 4$ continues the same positive lattice and is the reference bulk, so $E(4) = \eta$; it is not an abnormal interface. Gap $g = 6$ is the elementary interface of Section 4, and equation (6) gives $E(6) = c_6$.

For a finitely supported vector v, extend it by zero and compute its full range-two image. Self-adjointness gives the exact Rayleigh identity

$$\frac{\langle v, H_g v\rangle}{\langle v, v\rangle} = \frac{\|A_g v\|^2}{\|v\|^2}. \tag{8}$$

The certified upper endpoint in equation (4) yields

$$T := \frac{988671163952541}{125000000000000} = \frac{7905369311620328}{10^{15}} + \frac{1}{250} > c_6 + \frac{1}{250}.$$

It is enough, therefore, to exhibit a finite vector with quotient strictly larger than T.

5.2 Exact finite-gap comparisons

The positive abnormal gaps below nine other than six are

$$g \in \{1, 2, 3, 5, 7, 8\}.$$

For each row in the following table, a displayed finite-support integer vector in the supplementary witness table has $\|v\|^2 = D$ and $\|A_g v\|^2 = N$. The final column is the exact integer

$$\Delta(N, D) = 125000000000000\, N - 988671163952541\, D.$$

g	N/D	$\Delta(N, D)$
1	812/97	5598897096603523
2	866/109	484843129173031
3	3114/393	702232566651387
5	764/96	587568260556064
7	768/97	98897096603523
8	19672/2487	174815250030533

Every entry in the last column is positive. Hence every listed quotient is strictly larger than T, and equation (8) proves

$$E(g) > c_6 + \frac{1}{250} \qquad (g \in \{1, 2, 3, 5, 7, 8\}).$$

The smallest margin is the gap-eight value

$$\frac{19672}{2487} - T = \frac{174815250030533}{310875000000000000} > 0.$$

All vector images are evaluated on the support enlarged by two sites at each end, so no outgoing coordinate is omitted. The full vectors and image coordinates are placed in the supplement because they are direct integer reconstructions of the six rows above.

5.3 Uniform control of the large gaps

One fixed vector covers the infinite tail. On the consecutive interval $[-2, 11]$, take

$$v_* = (4, 0, 7, 8, 8, 9, 1, 7, 3, 6, 1, 4, 1, 2), \qquad \|v_*\|^2 = 391.$$

Only coefficients $\tau_i^{(g)}$ with $-4 \le i \le 11$ enter the full image on $[-4, 13]$. The right interface endpoint can therefore affect the calculation only when $g = 9$ or $g = 10$; every $g \ge 11$ has the same local coefficient word. Direct use of the four-term formula for A_g gives

$$\|A_9 v_*\|^2 = 3102, \qquad \|A_{10} v_*\|^2 = 3094, \qquad \|A_g v_*\|^2 = 3094 \quad (g \ge 11).$$

Consequently every $g \ge 9$ satisfies

$$E(g) \ge \frac{3094}{391} = \frac{182}{23}.$$

The remaining comparison is again exact:

$$\frac{182}{23} - T = \frac{10563229091557}{2875000000000000} > 0.$$

This proves the required strict separation uniformly on the entire large-gap tail. The three displayed squared norms, together with finite propagation, are a proof for all $g \ge 9$, not a sample of those gaps.

5.4 Reference and single-gap hierarchy

Theorem 5.1 (Complete abnormal single-gap hierarchy)**.** *For both τ-lifts and both interface orientations,*

$$E(4) = \eta < c_6 = E(6),$$

and, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

Thus G6 uniquely minimizes the squared spectral edge among abnormal positive single-gap interfaces.

Proof. The reference and G6 equalities come from Proposition 2.3 and equation (6); the strict inequality $\eta < c_6$ follows from the exact isolating intervals already displayed. Subsection 5.2 treats every positive gap below nine except 4 and 6, and Subsection 5.3 treats every gap at least nine. These sets exhaust the remaining positive integers. Lift negation preserves the squared spectrum, and reflection transfers the result to the opposite orientation. □

The theorem compares single abnormal positive gaps only. It makes no claim about arbitrary multi-gap configurations, interacting interfaces, or general finite cores. The constant 1/250 is a convenient certified uniform separation; no optimality is claimed for the size of that margin.

6 Phase Slips on Finite Rings

6.1 Legal residue constructions

We now close one, two, or three G6 interfaces on finite rings. Exponent notation 4^a denotes a consecutive reference gaps.

For $n = 8k + 2$, use

$$\mathcal{G}_2(k) = [6, 4^{2k-1}].$$

Its gap sum is $6 + 4(2k - 1) = 8k + 2$, it has $2k$ positive Q-sites, and its single interface has return distance $D_2(n) = n$.

For $n = 8k + 4$, use

$$\mathcal{G}_4(k) = [6, 4^{k-1}, 6, 4^{k-1}].$$

Its sum is $12 + 8(k - 1) = 8k + 4$, it again has $2k$ positive sites, and the two interface cores are separated by $D_4(n) = n/2$ in both directions.

For $n = 8k + 6$, put

$$a = \left\lfloor \frac{2k-3}{3} \right\rfloor, \qquad b = \left\lfloor \frac{2k-2}{3} \right\rfloor, \qquad c = \left\lfloor \frac{2k-1}{3} \right\rfloor$$

and use

$$\mathcal{G}_6(k) = [6, 4^a, 6, 4^b, 6, 4^c].$$

Writing $2k - 3 = 3u + v$, $v \in \{0, 1, 2\}$, verifies $a + b + c = 2k - 3$. Hence the gap sum is

$$18 + 4(a + b + c) = 8k + 6,$$

there are $2k$ positive sites, and the minimum core separation is

$$D_6(n) = 6 + 4 \left\lfloor \frac{2k-3}{3} \right\rfloor.$$

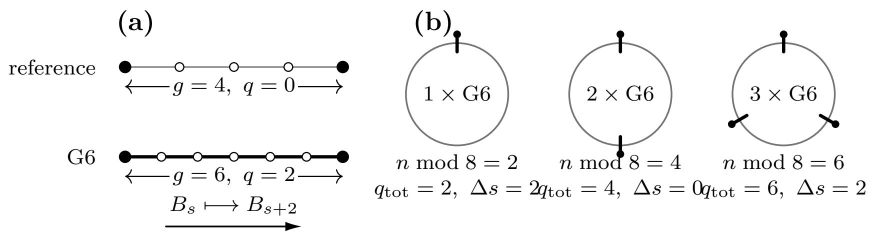

Figure 2: Reference gaps and charge-compatible phase slips. A G6 replaces $g = 4, q = 0$ by $g = 6, q = 2$; one, two, and three copies provide the mod-eight total charges for residues $2, 4, 6$, while their mod-four sector transport is checked independently.

These formulas are used only when their exponents are nonnegative.

In every construction the number of positive Q-sites and n are even. Thus

$$\prod_i Q_i = (-1)^{n-2k} = 1,$$

so both cyclic τ-lifts exist. The mod-eight charge check is separate: each G6 contributes $q = 2$, and the total charges $2, 4, 6$ agree with the three order residues. The translation-sector check uses Proposition 3.2: the total shifts are $2, 0, 2$ modulo four, equal respectively to n modulo four. After these two independent closure checks, either Hamilton holonomy $\alpha = \pm 1$ may be chosen.

6.2 Identification of local patches

Consider any of the preceding rings with $r \in \{1, 2, 3\}$ marked G6 cores, and let D be their minimum cyclic site separation; for $r = 1$, set $D = n$. Here D is measured in ring sites, not period-eight cells. Let a real cutoff be supported in a cyclic interval of radius R. Since $H = A^2$ has range four, its enlarged support has radius $R + 4$.

Proposition 6.1 (Finite-ring patch identification)**.** *Assume*

$$2(R + 4) < D.$$

Every enlarged localization patch is, after cyclic translation, optional reflection, and diagonal switching, a finite section of either the period-eight reference operator or the canonical bilateral G6 operator. This covers both τ-lifts, both interface orientations, both Hamilton holonomies, patches crossing the displayed seam, and cyclic wraparound.

Proof. The separation inequality ensures that an enlarged support meets at most one non-four gap. If it meets none, its positive Q-sites form one congruence class modulo four, so translation identifies the coefficient word with a reference sector B_s. If it meets one, translate the left endpoint of the gap to zero. The visible gaps are then four on both sides of one gap six, hence the word is the forward G6 model; reversing the cyclic coordinate gives the other orientation.

Each enlarged support is a proper arc. On such an arc, recursive diagonal switching makes every step-one edge positive. Two lifts of the same local Q-word differ by a common sign, removed after squaring by equation (2). The Hamilton holonomy can be placed on any chosen step-one seam. Move it outside every interface support; if a bulk patch crosses the initially displayed seam, the same proper-arc switching moves that sign outside the local compression. Unwrapping a proper arc before translation handles cyclic wraparound. These operations prove equality of all coefficients needed by the range-four quadratic form, not merely similarity of a central subword. □

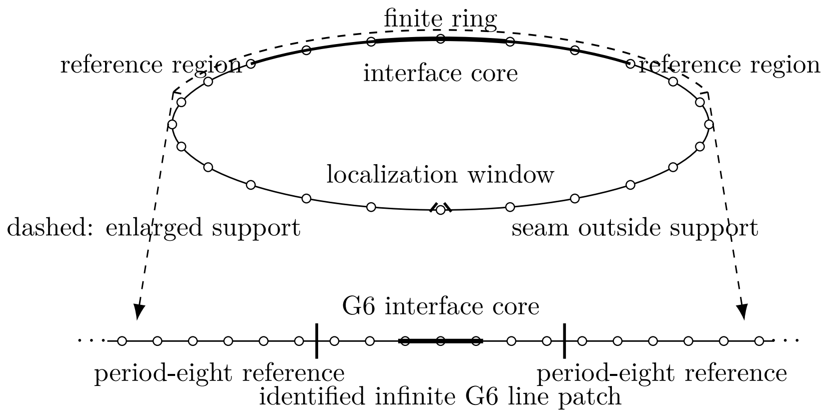

Figure 3: Local-to-global identification. A finite-ring cutoff and its range-four enlargement see either reference coefficients or one interface core; after unwrapping and switching, the latter is exactly a finite section of the bilateral G6 model. The holonomy seam is placed outside the interface support.

6.3 Discrete IMS localization

Let H be self-adjoint and let χ_j be real diagonal matrices with $\sum_j \chi_j^2 = I$. Entrywise,

$$[\chi,[\chi,H]]_{ab} = (\chi(a)-\chi(b))^2 H_{ab}.$$

For each pair a, b, the coefficient of H_{ab} in the following expression is one. Thus the exact discrete IMS identity is

$$H = \sum_j \chi_j H \chi_j + \frac{1}{2}\sum_j [\chi_j,[\chi_j,H]]. \tag{9}$$

For $R \geq 4$, define on the cycle

$$f_R(k) = \max\left(0, 1-\frac{\operatorname{dist}(k,0)}{R}\right), \qquad C_R = \sum_k f_R(k)^2 = \frac{2R^2+1}{3R},$$

and use every cyclic translate

$$\chi_j(a) = \frac{f_R(a-j)}{\sqrt{C_R}}.$$

If $n > 2R+4$, the supports do not self-collide in a range-four calculation, and $\sum_j \chi_j(a)^2 = 1$. For cyclic distance $d \leq 4$, direct summation gives

$$S_d(R) := \sum_j (\chi_j(a)-\chi_j(b))^2 = \frac{3(2d^2R - d(d^2-1))}{R(2R^2+1)}.$$

From equation (3), the absolute off-diagonal weights of H at distances $1, 2, 3, 4$ are bounded by $2, 1, 2, 1$. The following Schur row bound is a uniform operator-norm estimate for the IMS remainder:

$$2S_1 + S_2 + 2S_3 + S_4 = \frac{240R-342}{R(2R^2+1)} \leq \frac{120}{R^2}. \tag{10}$$

Proposition 6.2 (Separated-interface cap). *If $2(R+4) < D$, then every legal ring above satisfies*

$$\rho(A)^2 \leq c_6 + \frac{240R-342}{R(2R^2+1)} \leq c_6 + \frac{120}{R^2}.$$

Proof. By Proposition 6.1, every localized vector is unitarily identified with a reference or G6 line vector. Their squared spectral edges are $\eta < c_6$ and c_6, respectively, so

$$\langle \chi_j x, H\chi_j x\rangle \le c_6\|\chi_j x\|^2.$$

Sum over j, use $\sum_j \chi_j^2 = I$, and apply equations (9)–(10). Taking the supremum over unit vectors controls the full finite-ring spectral top. $\square$

6.4 Residue upper bounds and the analytic tail

For $s \in \{2,4,6\}$, the corresponding family has a fixed number of interfaces while $D_s(8k+s) \to \infty$. Choose

$$R = \left\lfloor \frac{D_s(8k+s) - 9}{2} \right\rfloor.$$

Then $2(R+4) < D_s(8k+s)$, and $R \to \infty$. Proposition 6.2 gives the one-sided conclusion

$$\limsup_{k\to\infty} m_{8k+s}^2 \le c_6, \qquad s \in \{2,4,6\}. \tag{11}$$

This proves neither a lower bound nor a limit or common liminf.

We next obtain an explicit analytic tail. The comparison threshold obeys

$$\theta_n > 8 - \frac{200}{n^2},$$

by $\cos x > 1 - x^2/2$ and $\pi^2 < 10$. Let

$$\mathcal{E}(R) = \frac{240R - 342}{R(2R^2+1)}.$$

At the first order in each residue class at or above 240, exact arithmetic gives:

n	certified squared cap	$8 - 200/n^2$
240	1561/200	2303/288
242	257368059342729114019/32519875000000000000	117078/14641
244	145320800767733426170/18296250000000000000	59511/7442
246	259114032812893881301/32412500000000000000	120982/15129

In every row the middle fraction is strictly smaller than the last by integer cross-multiplication. The first row uses the period-eight competitor; the other rows are the upper endpoint for c_6 plus $\mathcal{E}(R)$ for the displayed residue construction.

For $R \ge 4$,

$$\mathcal{E}(R) - \mathcal{E}(R+1) = \frac{6(160R^3 - 102R^2 - 262R - 171)}{R(R+1)(2R^2+1)(2R^2+4R+3)} > 0.$$

The numerator equals 7389 at $R = 4$, and its derivative is positive thereafter. Thus the strict decrease is analytic, not numerical. Along each residue subsequence, D_s and the chosen R do not decrease, so the spectral cap does not increase, whereas $8 - 200/n^2$ strictly increases. The four endpoint checks therefore propagate to every later order. We have proved

$$m_n < \rho_-(n) \qquad \text{for every even } n \ge 240.$$

This phase-slip argument explains why failure is eventually uninterrupted. It does not locate the sharp beginning at 48; that conclusion also uses the exact finite bridge in Section 7. Neither the exact-$2r$ count nor the rank-two multiplicity is used in this analytic tail.

6.5 Structural refinement for separated interfaces

Theorem 6.3 (Separated-interface cluster)**.** *Let a legal finite ring contain exactly* $r \in \{1, 2, 3\}$ *separated G6 interfaces, with either* τ*-lift, either orientations, and either Hamilton holonomy. If the minimum cyclic core separation satisfies* $D \geq 1040$*, then*

$$\operatorname{rank} \mathbf{1}_{[c_6 - 1/400,\, c_6 + 1/400]}(H) = 2r.$$

Eigenvalues are counted with multiplicity; no individual simplicity is asserted.

Each isolated G6 has a rank-two squared eigenspace, one mode from each simple unsquared partner. Truncating both modes at each interface gives $2r$ quasimodes. Exact tail and Gram bounds keep them independent and give the lower spectral count. On their orthogonal complement, patch identification, the single-interface complement bound, and discrete IMS give the matching upper count. The full Gram, complement, and effective-operator estimates belong to the separate supplement. Theorem 6.3 is a structural refinement of the separated-interface picture. It is not used in Proposition 6.2, the analytic $n \geq 240$ tail, or the exact argument placing the onset of continuous failure at 48.

7 Finite Completion of the Classification

7.1 Exact-computation protocol

7.2 Orders 8 through 32

7.3 Orders 34 through 46

7.4 The exact bridge from 48 to 238

7.5 Exhaustive synthesis

8 Concluding Remarks

Data and Code Availability

A Exact Spectral Certification of the Reference Phase and G6

B Exact Finite Classification and Completeness

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